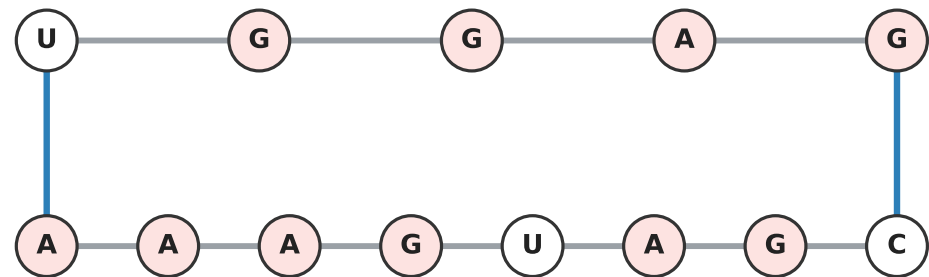


Finding a kink-turn: candidate retrieval and motif-specific validation

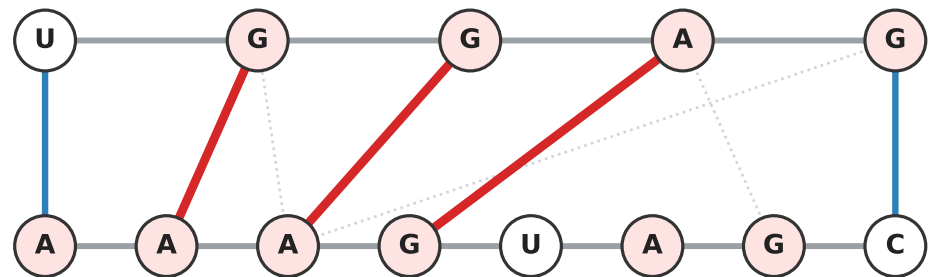
(a) Watson-Crick pairs only

an ordinary asymmetric internal loop



(b) + non-WC edges: tandem sheared G·A

candidate kink-turn signature



— Watson-Crick pair — sheared G·A (Saenger XI) other non-WC contact — backbone

(c) Confirmed representatives

PDB	Functional context	Sheared G·A pairs (Saenger XI)	Strands (nt)
2GIS	SAM-I riboswitch (anchor)	A18-A37, A19-A36, A20-A35	5 / 8
1E7K	U4 snRNA · 15.5K protein	D32-D44, D33-D43	4 / 7
4BW0	HMKt-7 · L7Ae protein	A4-A22, A5-A21, A6-A20	5 / 8
2OZB	U4 snRNA · Prp31-15.5K-U4 snRNA complex	F32-F44, F33-F43	4 / 7

Sheared G·A geometry (Saenger XI) evaluated with MC-Annotate; graph topology alone does not encode it.